

Polyethylene terephthalate microplastics affect gut microbiota distribution and intestinal damage in mice

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ABSTRACT

Polyethylene terephthalate microplastics (PET-MPs) have been detected in the environment and human metabolites or tissues; however, their potential effects on humans under actual exposure doses remain unclear. Herein, male adult mice were exposed to 10 μ m PET-MPs at concentrations of 10, 50, and 250 mg/kg per body weight consecutively for 28 days. Changes in blood biochemistry, inflammatory factors, colonic histopathology, colonic mucus gene mRNA levels, and the gut microflora were monitored to study PET-MPs toxicity. The results showed that PET-MPs exposure increased relative serum alanine aminotransferase (ALT) and glucose (GLU) levels in 50 mg/kg bw PET-MPs exposure group, and altered relative levels of inflammatory factors, thereby inducing the inflammatory response. Moreover, PET-MPs exposure increased mRNA expression levels of colonic mucus secretion related and barrier function related genes, indicating intestinal mucus secretion and barrier integrity dysfunction, which was consistent with the results of histopathological results. In addition, gut microbiota analysis revealed that the diversity and community composition were altered after PET-MPs exposure, suggesting a metabolic disorder. Therefore, our results demonstrated that exposure to PET-MPs led to intestinal injury and changes in the gut microbiome composition in mice. Overall, the study findings provided basic data about the health risks of PET-MPs to humans, highlighting that MPs-induced toxicity warrants more concern in the future.

1. Introduction

Plastics are used worldwide for their outstanding properties, such as durability and low price. However, plastics are generally nondegradable and typically crack into finer particles under physicochemical or biological processes persisting for decades. Microplastics (MPs) are defined as plastic particles less than 5 mm in diameter (Wang et al., 2021). Consistent with plastics production, common MPs types mainly include polyethylene MPs (PE-MPs), polypropylene MPs (PP-MPs), polystyrene MPs (PS-MPs), polyethylene terephthalate MPs (PET-MPs), and polyvinyl chloride MPs (PVC-MPs). Among plastics, PET represents a considerable portion of plastic production due to its valuable properties, such as being lightweight, ease of molding, and low cost, and it is commonly employed in food and beverage containers (Grogan et al., 2021; Pencik

et al., 2023). With huge quantities produced, the prevalence of PET leads to inevitable human exposure to PET-MPs in daily life. Research has revealed that PET-MPs pollutants are present in marine (Li et al., 2022a) and river water, sediments (Islam et al., 2022; Li et al., 2020a), and aquatic organisms (Devereux et al., 2021; Gao et al., 2023), and in the dust (Liu et al., 2019), and consumed food or water, such as sea salt (Seth and Shrivastav, 2018; Yang et al., 2015), beverages (Li et al., 2022b), bottled water (Praveena et al., 2022) or drinking water (Pivokonsky et al., 2018). Moreover, PET-MPs, as the predominant MPs type, have been detected in human feces (Yan et al., 2022), blood (Leslie et al., 2022), and lung tissues (Jenner et al., 2022).

Due to their environmental ubiquity and persistence, humans may ingest MPs through various tracts (Senathirajah et al., 2021). The main routes of human exposure to microplastics have been identified as

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ingestion, inhalation and dermal contact, with ingestion considered to be the main route of human exposure (Xu et al., 2022; Zarus et al., 2021). The intestine, as the first organ confronted with microplastic ingestion, is likely to be the main affected organ. With functions in digestion, absorption, and immunity, the intestine plays a critical role in maintaining normal physiological growth and development of the host. Inflammation is an innate defense mechanism against injury, infection, and stress. Adequate inflammatory stimulation is beneficial to the host, but excessive inflammation then leads to tissue damage (Groschwitz and Hogan, 2009). The intestine, as an important immune defense of the body, can be stimulated by inflammation-causing disruption of the tight junction barrier, which in turn triggers intestinal inflammation and disease development (Wu et al., 2020). In addition, the intestine contains abundant and diverse microbial communities, which are described as the gut microbiota (Aidy et al., 2013). Harboring the second largest genome of the host, the composition and homeostasis of the gut microbiota forms a complex and dynamic system that interacts with host physiological functions (Li et al., 2022c), and is even involved in the pathogenesis of irritable bowel syndrome, obesity-related disease, liver disease, and cancer (Marchesi et al., 2016). In this aspect, researchers have been concerned about the MPs exposure-induced health effects on the digestive system. According to previous studies, microplastic intake may cause inflammation in aquatic organisms such as zebrafish and even lead to intestinal damage, which is closely associated with intestinal microbiota imbalance or metabolism dysregulation. Accumulating attempts have investigated the interconnections between the gut microbiota and microplastic ingestion in mice, such as PS-MPs (Jin et al., 2019; Lu et al., 2018), PE-MPs (Li et al., 2020b; Sun et al., 2021), and PVC-MPs (Chen et al., 2022a,b). PET, as a type of polymer plastic commonly used to manufacture plastic bags, bottles, and clothing. In addition, PET-MPs have been widely detected in the environment and human tissue samples. Despite the previous research has shown that PET-MPs induce sublethal toxic effects in zebrafish early-development (Teodoro et al., 2024), the toxicity and mechanism of PET-MPs in mammals remains limited and unclear.

The objective of this study was to gain an understanding of the effects of PET-MPs on the intestinal, including the inflammatory response, gut microbiota, and intestinal barrier function after 28 days of continuous exposure using C57BL/6 mice as a mammalian model. The special features of our study include the selection of an understudied MP, focus on the dose effects from actual environmental exposure concentrations, and observation of two time periods to explore temporal effects on the gut microbiota. The study findings may provide a new toxicity assessment for the potential health risks of MPs on humans.

2. Materials and methods

2.1. Microplastics

The PET-MPs (diameter: 1–20 μm , density: 1.37 g/cm^3) used in the study were obtained from Tesulang Chemical Material Co., Ltd (Dongguan, China) and had been characterized by scanning electron microscopy (SEM, JEOL JSM-7900F, Japan) and I-Raman plus spectroscopy system (I-Raman, BWS465–785S, B&W Tek, USA).

2.2. Animals and treatment

Forty six-week-old male C57BL/6 mice (weighing 18–20 g) were purchased from the Laboratory Animal Center, Nanjing Medical University, Nanjing, China. All the mice were housed in the animal room under a 12 h light/ 12 h dark cycle with food and water available ad libitum. The ethical approval for the animal experiment was obtained from the Institutional Animal Care and Use Committee of Nanjing Medical University (IACUC-2210016), approved on 11 October 2022.

After acclimating for 7 days, all the mice were randomly divided into four groups ($n = 10$) and received PET-MPs (10, 50, and 250 mg/kg bw)

or an equal volume of olive oil by oral gavage once daily for 28 days, respectively. The PET-MPs concentrations were selected according to environmentally realistic concentrations of MPs in rivers and levels to which humans are exposed via consumer products. The dry PET-MPs powder were dispersed in olive oil and subjected to supersonic wave vibration for 20 min to ensure full dispersion before oral gavage in mice. During the experiment, we replaced the padding in the cages and then immediately collected enough fresh feces from each group on days 14 and 28, which were stored at -80°C . All the mice were fasted for 8 h, anesthetized with pentobarbital sodium, and sacrificed after 4 weeks of exposure, and then the tissues and blood samples were quickly collected.

2.3. Histopathology

Colon tissues were collected from 3 mice in each group and immediately fixed with 4 % paraformaldehyde solution for 48 h, dehydrated in a graded ethanol series, hyalinized in xylene, and embedded in paraffin. Then, 3 distal colon samples from each group were cut into 5 μm -thick sections and stained with hematoxylin and eosin (H&E). Meanwhile, a 5 μm section of the middle colon was stained with Alcian blue-periodic acid Schiff (AB-PAS) solution. Subsequently, the histopathological changes in colon tissues were examined using a BX53 microscope (Olympus, Japan).

2.4. Biochemical analysis

After the PET-MPs exposure experiments, the peripheral blood samples were centrifuged at 4°C and $3000 \times g$ for 15 min to extract the serum. Colon tissues were homogenized with precooled phosphate-buffered saline (PBS) solution ($\text{pH}=7.2\text{--}7.4$) for protein extraction, and then centrifuged the homogenate to obtain the protein supernatant for analysis.

Hematological indices including white blood cell (WBC) and red blood cell (RBC) counts and hemoglobin (HGB) levels in blood were measured using the fully automated hematology analyzer (ADVIA2120i, Germany). Blood biochemical indices, such as alanine aminotransferase (ALT), aspartate transaminase (AST), glucose (GLU), and triglyceride (TG) levels in the serum were determined using the fully automatic biochemical analyzer (7100, Germany) according to the manufacturer's instructions. The $\text{TNF-}\alpha$, IL-2, IL-5, and IL-10 contents in the serum and colon tissues were measured using ELISA kits (Jiangsu, China) in accordance with the manufacturer's instructions.

2.5. Colon RNA extraction and gene expression analysis

Total RNA from the colon was extracted with TRIzol reagent (TRANS, Beijing, China) according to the manufacturer's instructions, and the quantity and quality of RNA were examined with a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific, USA). The reverse transcription system using PrimeScript® RT Reagent Kit (TaKaRa, Japan) was applied to transcribe mRNA to cDNA. For RT-qPCR, a 7900AB fluorescence quantitative PCR instrument (ABI, USA) was employed to measure the mRNA expression levels of *Muc2*, *Muc3*, and *claudin1* using SYBR Green Supermix (Vazyme, Nanjing, China). The β -actin gene was used as an internal reference gene. All primers used in this study were listed in Table S1.

2.6. Fecal DNA extraction, qPCR amplification, and 16S rRNA gene sequencing

All mouse fecal DNA extraction procedures were performed using the E.Z.N.A.® DNA kit (Omega BioTek, USA) according to the instructions. The V3–V4 region of 16S rRNA gene was amplified by the PCR system using the primers 338 F (5'-ACTCCTACGGGAGGCAGCAG-3') and 806 R (5'-GGACTACHVGGGTWTCTAAT-3'). The amplification products were detected by 1 % agarose gel electrophoresis, and then were quantified

followed by homogenization to constitute the sequencing library. The original sequenced sequences were spliced using FLASH and the sequences were clustered into operational taxonomic units (OTUs) based on 97 % similarity. Alpha and beta diversity, and species composition analysis were analyzed on the online platform of Majorbio Cloud Platform (www.majorbio.com) to evaluate alterations in the gut microbiota.

2.7. Statistical analysis

All data are presented as the mean $\pm$ standard error (SD). Statistical analysis was performed using GraphPad Prism 9. Data normality and homogeneity of variance were verified using the Kolmogorov-Smirnov one-sample test and the Levene test. One-Way ANOVA analysis was used to analyze multiple sets of continuous data when the data met the criteria for normality and homogeneity of variance, followed by the Dunnett test to assess between-group differences. Otherwise, the Kruskal-Wallis test was employed, and the data were adjusted using the Bonferroni correction. $P < 0.05$ was considered as statistically significant.

3. Results

3.1. Characterization of PET-MPs

The morphology of PET-MPs identified by the SEM was irregular in shape and the particle surface was relatively rough (Fig. 1A). Further analysis showed that the PET-MPs diameters ranged from 4 to 20 μm and was concentrated between 8 and 14 μm (Fig. 1B). Consistent with PET references in previous research (Jin et al., 2022; Mei et al., 2022), the Raman spectrum of PET-MPs showed typical peaks of PET-MPs at 631, 854, 1282, and 1613 cm^{-1} (Fig. 1C).

3.2. Effects of PET-MPs exposure on body weight and organ coefficient

The body weights tended to decrease when the mice were exposed to high doses of PET-MPs (50 and 250 mg/kg bw). The average heart, spleen, and testis weight relative to body weight in PET-MPs treated groups were decreased compared to those in the control group. However, these differences were not statistically significant (Table S2).

3.3. Effects of PET-MPs exposure on blood biochemistry

The blood hematological indexes showed that the percentage of hematocrit (HCT) in the 50 mg/kg bw PET-MPs group was significantly higher than that in the control group ($P < 0.05$). Regarding the results of biochemical tests, the ALT and GLU levels in the 50 mg/kg bw PET-MPs group were significantly higher than those in the control group ($P < 0.05$ and $P < 0.01$, respectively). In particular, the serum ALT level in the 250 mg/kg bw PET-MPs group was significantly lower than that in the 50 mg/kg bw PET-MPs group ($P < 0.05$). AST, total protein (TP),

TG, and cholesterol (CHOL) levels did not differ significantly among groups (Table S3).

3.4. Effects of PET-MPs exposure on inflammatory factors in the serum and colon

The relative levels of inflammatory factors in the serum and colon were shown in Fig. 2A and B, respectively. In the serum, the relative levels of IL-10 were significantly decreased and those of IL-5 were significantly increased after exposure to PET-MPs compared to the control. The IL-2 and TNF- α relative levels also showed upward trends compared to the control, which were significantly increased after exposure to 50 and 250 mg/kg bw PET-MPs. Similarly, compared with the control group, the relative levels of IL-2 and TNF- α in the colon were significantly higher and those of IL-10 were significantly lower among all PET-MPs treated group.

3.5. Effects of PET-MPs exposure on colon histopathology

The H&E staining of the colon showed the accumulation of inflammatory cells (black arrow), slightly damage to the colonic crypt structure (black border pentagram) and mucosal epithelial (blue triangle), partial dilation of glands (red triangle), and thinning of the muscle layer (black line) to varying degrees after PET-MPs exposed. Especially, lymphoid follicles were observed in the 250 mg/kg bw PET-MPs group (black arrow) (Fig. 3A). AB-PAS staining revealed that mucus secretion was reduced in the PET-MPs groups (Fig. 3B). Further quantitative analysis showed that the proportion of colonic mucus secretion was significantly reduced in the 250 mg/kg bw PET-MPs group ($P < 0.05$) (Fig. 3C).

3.6. Effects of PET-MPs exposure on mRNAs related to colonic mucus secretion

The relative mRNA levels of *Muc1*, *Muc2*, *Muc3*, *Klf4*, *Occludin*, and *Claudin-1* in the colon of the PET-MPs treated groups showed downward trends compared to the control group, although there was no significant difference among groups in the mRNA levels of *Claudin-1*. Especially, the *Muc2* mRNA levels were significantly decreased in all PET-MPs treated groups. (Fig. 3D)

3.7. Effects of PET-MPs exposure on gut microflora

The 16S rRNA sequences with at least 97 % similarity were clustered into operational taxonomic units (OTUs) to analyze the gut microbiota in mouse feces. According to the richness and diversity indexes, alpha diversity did not significantly differ between the PET-MPs treated and control groups after 14 consecutive days of PET-MP exposure. At 28 days, the Shannon index of the 250 mg/kg bw PET-MPs group was significantly decreased compared to that of the control group

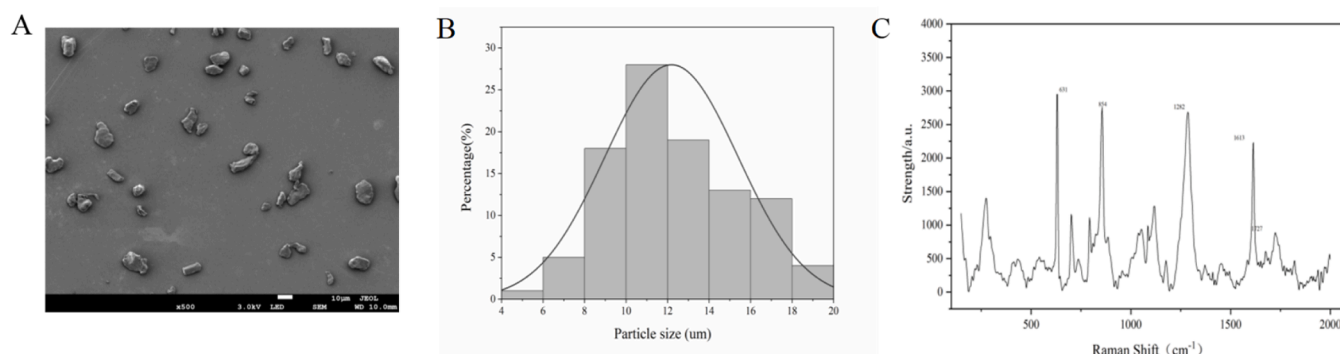


Fig. 1. Characterization of the PET-MPs used in this study. (A). SEM image of PET-MPs; (B). Distribution of PET-MPs diameters; (C). Raman spectrogram of PET-MPs.

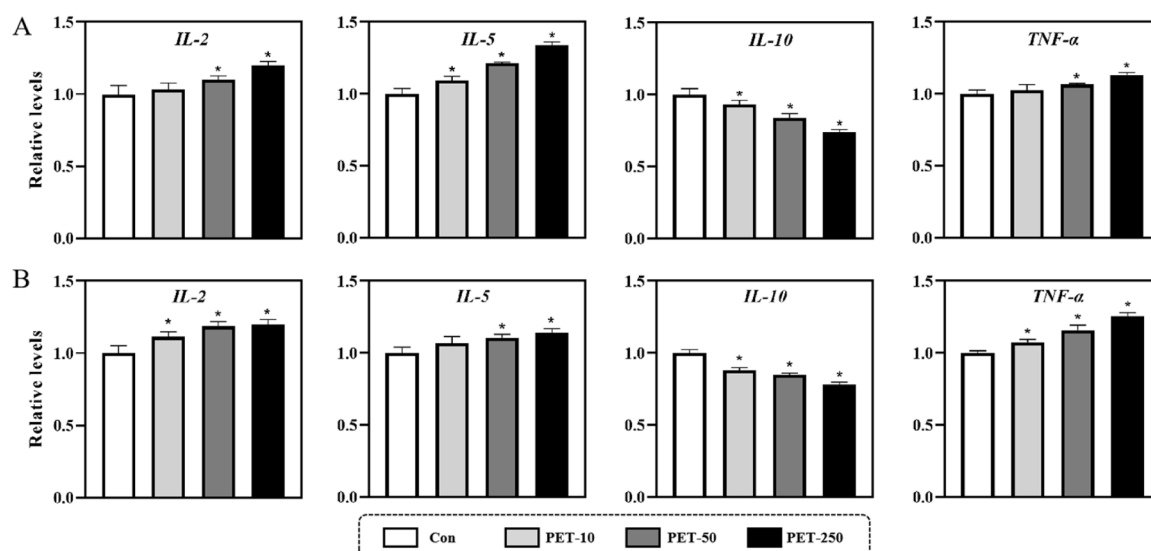


Fig. 2. Effects of PET-MPs exposure on inflammatory factors in the serum (A) and colon (B), followed as: IL-2, IL-5, IL-10, and TNF-α. Four groups were 0, 10, 50, and 250 mg/kg bw in order. * $P < 0.05$.

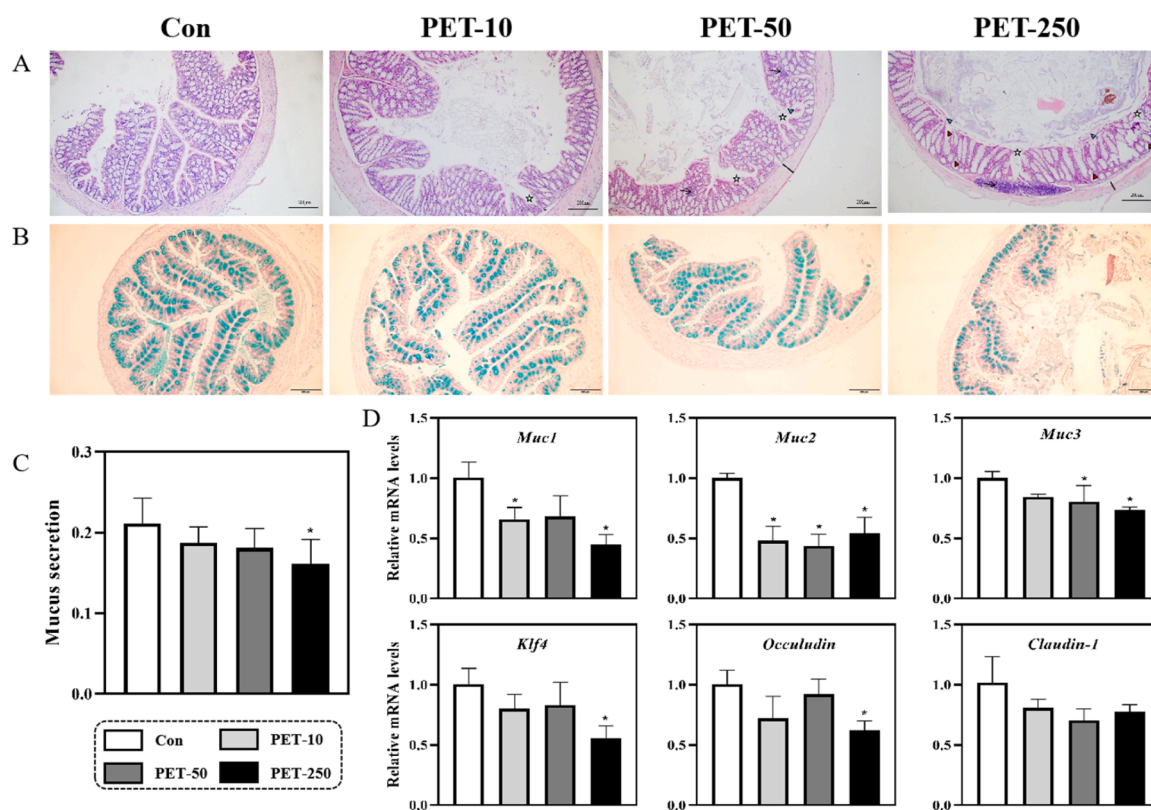


Fig. 3. Effects of PET-MPs exposure on the mucus secretion and intestinal barrier dysfunction in the gut of mice. (A). H&E staining of the colon; (B). AB-PAS staining of the colon; (C). The ratio of the mucus secretion area to the entire colon area; (D). The relative mRNA expression levels of mucus secretion and intestinal barrier. Black arrow: inflammatory cells accumulated; Black border pentagram: colonic crypt structure damaged; Blue triangle: mucosal epithelial damaged; Red Triangle: glands dilated; Black line: muscle layer thinner. Four groups were 0, 10, 50, and 250 mg/kg bw in order. * $P < 0.05$.

($P = 0.0406$) (Fig. 4A). With respect to beta diversity, the differences in the bacterial composition among groups were visualized by principal coordinates analysis (PCoA). The PCoA results showed that the microbial composition of the PET-MPs treated groups was notably different from that of the control group (Fig. 4B). In addition, the Venn diagrams showed the common and unique OTUs among all the groups after PET-MPs exposure for 14 days and 28 days, respectively, with 390 OTUs

shared among all groups at 14 days, which slightly increased to 392 OTUs at 28 days (Fig. 4C).

The changes in the gut microbiota composition at the phylum and genus levels after PET-MPs exposure were shown in Fig. 5. In brief, Bacteroidota and Firmicutes were the main phyla, followed by Actinobacteriota, Verrucomicrobiota, and Proteobacteria. (Fig. 5A). In addition, *norank_f_Muribaculaceae*, *Lactobacillus*, *Lachnospiraceae_NK4A136*

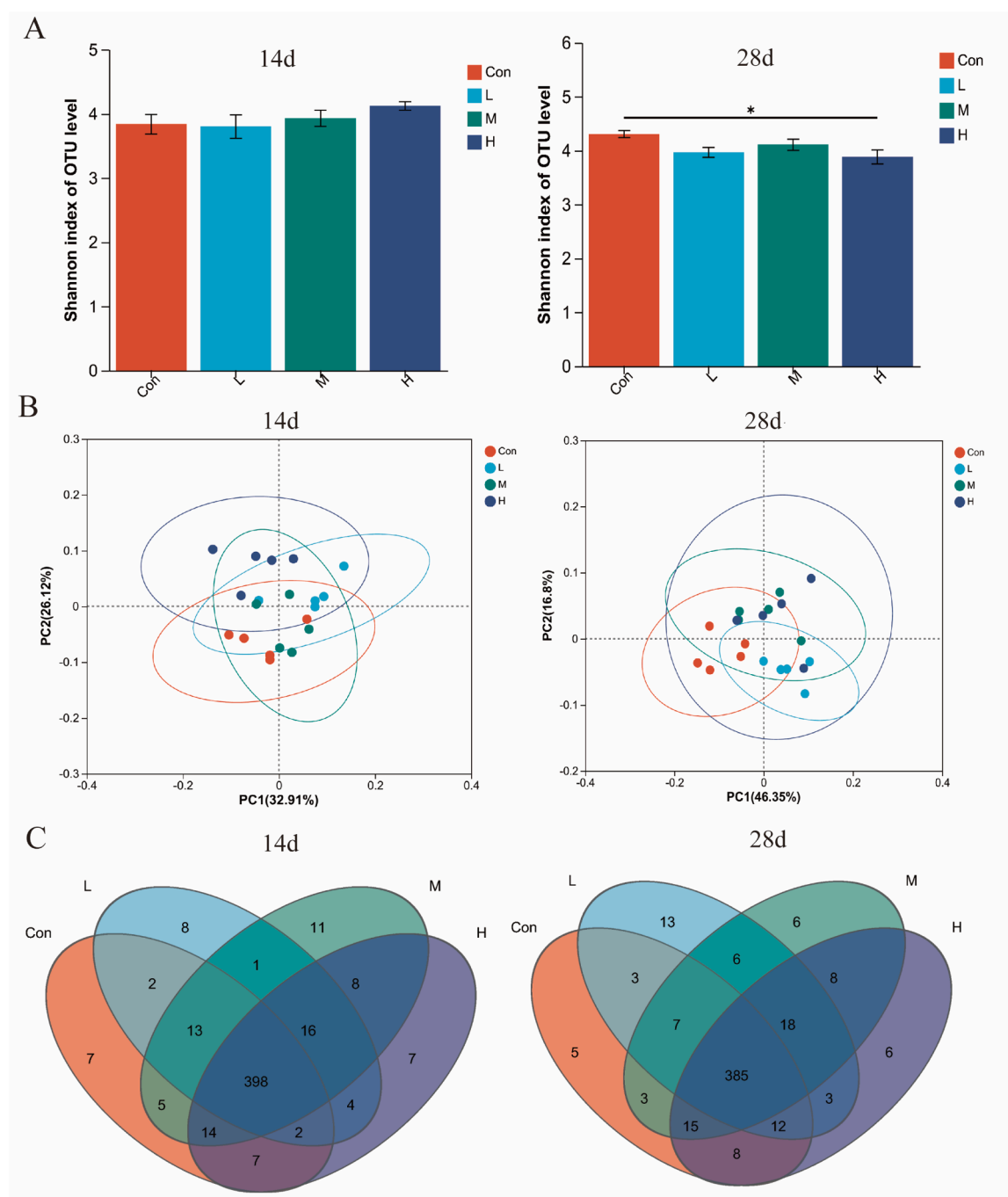


Fig. 4. Effects of PET-MPs exposure on the diversity of gut microbiota in the feces of mice. (A). Shannon index in the gut microbiota after PET-MPS exposure for 14 days and 28 days, respectively; (B). Principal coordinates analysis (PCoA) of the gut microbiota from the control and PET-MPs treated groups for 14 days and 28 days, respectively; (C). Change in operational taxonomic units (OTUs) in the gut microbiota after PET-MPs exposure for 14 days and 28 days, respectively. Four groups were 0, 10, 50, and 250 mg/kg bw, named Con, L, M, H in order. * $P < 0.05$.

group, *Bifidobacterium*, and *norank_f_norank_o_Clostridia_UCG-014* were the main genera. (Fig. 5B) After the application of PET-MPs, the composition of gut microbiota was changed. At the phylum level, the relative abundance of Bacteroidota was significantly reduced in the feces of the PET-MPs treated groups after exposure for 14 days ($P = 0.0060$) compared to the control group, while the relative abundance of Firmicutes ($P = 0.0022$) and Actinobacteriota ($P = 0.0066$) were significantly increased. After PET-MPs exposure for 28 days, the relative abundance of Bacteroidota ($P = 0.0307$) and Actinobacteriota ($P = 0.0009$) were significantly increased, while that of Firmicutes was significantly decreased ($P = 0.0052$) (Fig. 5C). At the genus level, for

example, a lower *norank_f_Muribaculaceae* was recorded in the PET-MPs treated groups compared to the control group at 14 days, although the difference was not statistically significant. Meanwhile, the relative abundance of *norank_f_Muribaculaceae* was significantly increased in the PET-MP-treated groups at 28 days ($P = 0.0444$). (Fig. 5D)

4. Discussion

With the widespread detection of MPs in the environment media such as in water, soil, air, and even human metabolites and tissues, an increasing number of studies have recently focused on the potential risks

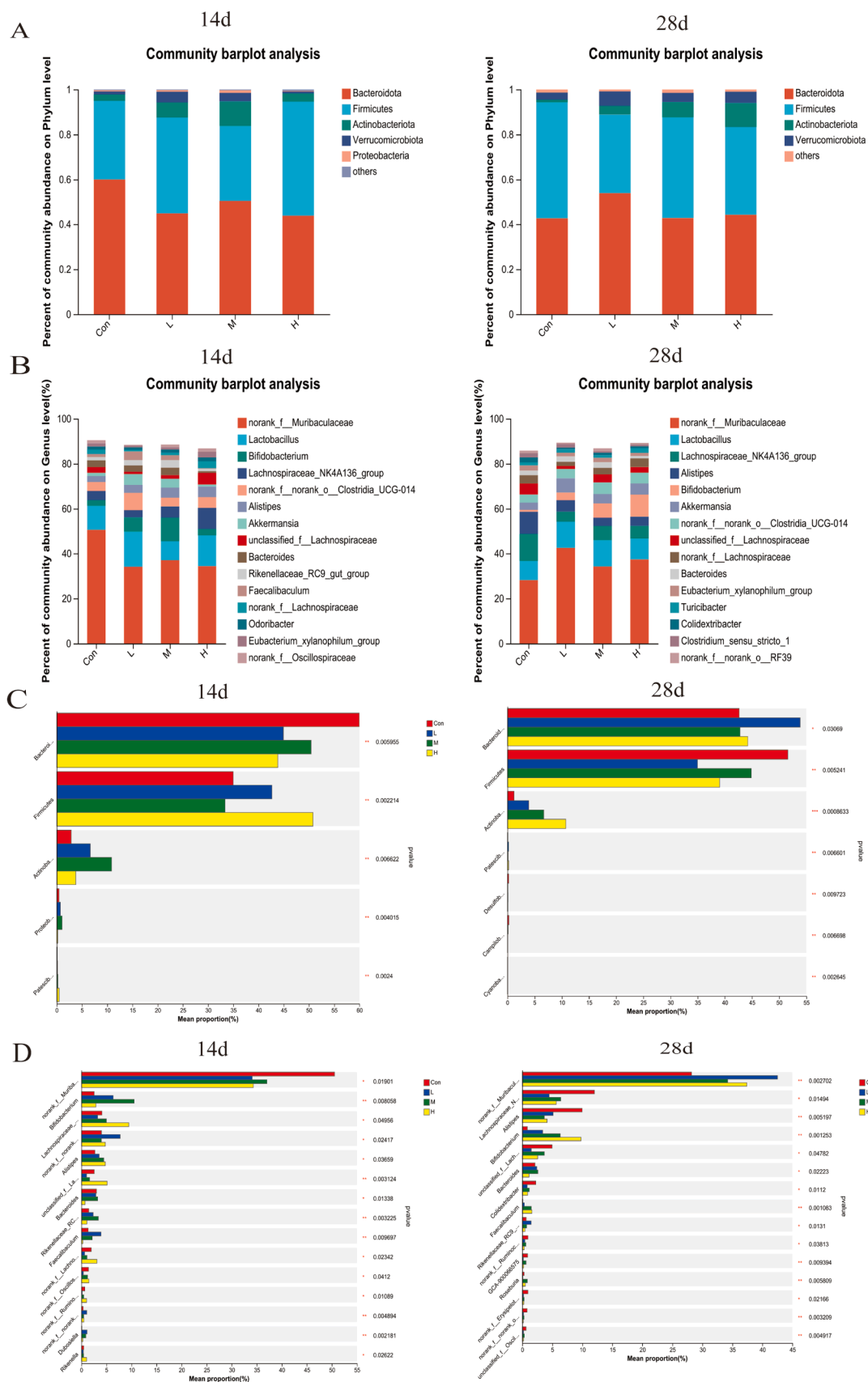


Fig. 5. Effects of PET-MPs exposure on the gut microbiota composition in the feces of mice. Change in the gut microbiota composition at the phylum (A), and genus (B) levels after PET-MPs exposure for 14 days and 28 days, respectively; Difference analysis in the gut microbiota composition at the phylum (C), and genus (D) levels after PET-MPs exposure for 14 days and 28 days, respectively. Four groups were 0, 10, 50, and 250 mg/kg bw, named Con, L, M, H in order. * $P < 0.05$.

of MPs to human health. The absorption of MPs by the human body occurs through various pathways, among which ingestion is considered to be an important route (Prata et al., 2020). Although previous studies have identified that common types of MPs, such as PE-MPs and PS-MPs, induce pathological changes in intestinal tissue, the effects of PET-MPs on intestinal mucin secretion and barrier function, as well as the specific mechanisms involved, remain unclear. The minimum PET-MPs concentration employed herein was based on the estimated mass content of MPs ingested by the human body through various pathways. It was estimated that human can ingest 0.1–5 g of microplastics per week (Senathirajah et al., 2021), which was converted according to the body weight of each adult was 60 kg, resulting in the exposure dose of microplastics in the real environment is 1.17–22.4 mg/kg bw. Based on previously established animal models, the exposure concentration of PS-MPs in mice is approximately 0.05–2.9 mg/day (Pan et al., 2024). Thus, the lowest exposure concentration of PET-MPs in mice was chosen as 10 mg/kg bw, with a dietary concentration of 0.2 mg per day considered to be an effective reflection of actual exposure level. In order to evaluate potential health issues in mice, we attempted to investigate the effects of PET-MPs exposure on the intestinal mucosal barrier function and gut microbiota composition in mice at three dose concentrations, with the doses of 50 and 250 mg/kg bw as medium and high dose groups to investigate the potential toxic effects of PET-MPs.

Pro-inflammatory cytokines including IL-2, IL-5, and TNF- α , and anti-inflammatory factors, like IL-10 were commonly involved in the occurrence and development of the inflammatory response and played a key regulatory and mediating roles in the overall immunomodulatory process (Dinarello and Mier, 1986; Lan et al., 2008; Ouyang et al., 2011). The expression of inflammatory cytokines in the serum and colon tissues of mice after PET-MPs exposure was investigated, and the prominent alteration of inflammatory factors confirmed a mild inflammatory response in the colonic epithelium. This result was further reinforced by the histopathological manifestation of inflammatory cell infiltration, suggesting that PET-MPs could act as foreign objects to trigger an inflammatory response in the host body. Inflammation might directly influence intestinal health, and the continuous development of intestinal inflammation could induce bacterial translocation, intestinal barrier dysfunction, and water–electrolyte imbalance (Bradley, 2008). Histological analysis demonstrated remarkable alterations in the colonic architecture in PET-MPs treated mice, as evidenced by the destruction of the crypt structure, damage to the mucosal epithelium, a thinned muscle layer, and even lymphoid follicles. These above observations indicated that PET-MPs at a relatively high dose may cause an inflammatory response and induce remarkable colon injuries.

In corroborating the colonic mucin content, we further utilized by AB-PAS staining of the colon tissue sections and scored. A downward trend in colonic mucin proportion was observed in the PET-MPs exposure groups, with the high-dose (250 mg/kg bw) PET-MPs group showed a significant decrease compared to the control group. Meanwhile, the results indicated that PET-MPs treatment could downregulate the transcriptional expression levels of key genes related to mucus secretion and barrier function, including *Muc1*, *Muc2*, *Muc3*, *Klf4*, and *Occludin*, which was consistent with previously reported results (Chen et al., 2022b; Jin et al., 2019; Lu et al., 2018). According to previous studies, *Muc2* was the main component of mucin secretion in the colon (Sun et al., 2021), while *Occludin* and *Claudin-1* also played important roles in maintaining epithelial barrier integrity (Ahn et al., 2016). Herein, the significantly decreased expression of *Muc2* suggested dysfunction of intestinal mucus secretion after exposure to PET-MPs, and the decreased expression of *Occludin* and *Claudin-1* indicated that PET-MPs exposure might disrupt the intestinal barrier, facilitating bacterial invasion (Chen et al., 2022b). Under normal circumstances, a complete intestinal barrier acts as a protective system to separate organisms from exogenous substances (Sussarellu et al., 2016). However, disruption of intestinal barrier balance might lead to increased intestinal permeability, which in turn can accelerate the translocation of pathogens or exogenous harmful

substances to the bloodstream, causing serious damage (Burcelin et al., 2012). Given the findings of previous work, impaired barrier function was considered to be a key determinant of intestinal susceptibility to inflammation (Chiocchetti et al., 2019). Thus, the results indicate that short-term exposure to a high-dose of PET-MPs can stimulate the inflammatory response and might further decrease mucin secretion, leading to impairment of the intestinal barrier function and increased the risk of intestinal diseases in the body.

In the analysis of the gut microbiota, differences in the gut microbiota were observed after PET-MPs exposure. After PET-MPs exposure for 14 days, the relative abundance of Bacteroides significantly decreased and that of Firmicutes significantly increased only in the 250 mg/kg bw PET-MPs group compared with the control group. However, different phenomena were observed after two more weeks of continued PET-MP exposure, with the relative abundance of Bacteroides increasing and that of Firmicutes decreasing. The gut microbiota composition in living organisms is a continuous and dynamic process, and changes in its community structure have been shown to lead to host metabolic disorders and even be related to the occurrence of certain diseases (Thaiss et al., 2016). After short-term exposure to PET-MPs, the diversity of gut microbiota in mice decreased, and structural changes such as an increase in the proportion of Firmicutes/Bacteroidetes (Xie et al., 2022). Harusato et al. found that PET-MPs with various morphologies might slightly affect the composition of the microbial community in mice according to the 16 s rRNA metagenomic analysis (Harusato et al., 2023). In addition, (Zhao et al., 2023) exposed SD rats to 10 mixed MPs including PET-MPs for 42 days, the abundance of Firmicutes increased from 54 % to 79 %, and Bacteroidetes decreased from 42 % to 16 % compared to control group. Firmicutes and Bacteroidetes typically account for over 90 % of the gut microbiota in healthy human adults, and the proportion of Firmicutes to Bacteroidetes was considered to be related to energy intake and even health conditions (Simpson and Campbell, 2015). An increasing in the Firmicutes to Bacteroidetes ratio might influence energy absorption or storage by reducing the ability to ferment polysaccharides, and even promote the possibility of obesity (Ley et al., 2005). Thus, the results suggested that energy intake in the intestine might increase after a short exposure time of PET-MPs and be reduced with prolonged exposure. In addition, the relative abundance of Actinobacteria increased with PET-MPs dosage, showing a significant increase at 28 days. According to previous studies, Actinobacteria participated in digestion and absorption in the human body (Adelman et al., 2020), suggesting that energy consumption might be enhanced after PET-MPs exposure. At the genus level, several microbial communities related to host health underwent significant changes induced by PET-MP exposure. For example, the relative abundance of *norank_f_Muribaculaceae* was significantly reduced in all PET-MPs treated groups at 14 days, while that of *Bifidobacterium* showed a significant increase only in the 50 mg/kg bw PET-MPs group. At 28 days, the relative abundance of *Bifidobacterium* was significantly increased with increasing PET-MP dosage, while that of *norank_f_Muribaculaceae* was significantly increased only in the 10 mg/kg bw PET-MPs group. Notably, *norank_f_Muribaculaceae* belongs to the family of probiotic bacteria associated with production of short-chain fatty acids (SCFAs) (Al-Bulish et al., 2022; Liang et al., 2021), which plays a crucial role in the regulation of energy metabolism and maintenance of intestinal mucosal barrier homeostasis (Morrison and Preston, 2016). The increased abundance of *norank_f_Muribaculaceae* was beneficial for improving gut structure and inflammation (Shang et al., 2021; Yu et al., 2021), while *Bifidobacterium* could exert anti-inflammatory effects by inducing intestinal IL-10 expression and preventing Th1-driven inflammation (Mohamed et al., 2021). The disruption of the gut microbiota might be caused by the consumption of MPs or chemical and mechanical disturbances within the gastrointestinal tract (Fackelmann and Sommer, 2019), which might further affect the immune system and metabolic functions of the host (Gao et al., 2025). These findings from this study have significant implications for enhancing the understanding

of the health risks that MPs pose to the human gut microbiome.

Serum ALT level was an important indicator for evaluating the degree of liver injury. Blood biochemistry analysis of mice showed that the serum ALT level increased with PET-MP exposure, suggesting potentially impaired liver function. Chen et al. reported a similar upregulation of ALT expression in serum after chronic exposure to PVC-MPs (Chen et al., 2022a). The liver, as one of the most metabolic organs, contributes to GLU regulation and maintaining lipid homeostasis (Frazier et al., 2023). Additionally, compared to the control group, the blood GLU concentration was significantly increased in the 10 mg/kg bw PET-MPs treated group ($P < 0.05$), indicating pronounced fluctuations in mouse GLU metabolism of mice after PET-MPs exposure. Fan et al. also observed elevated plasma GLU levels and insulin resistance after oral administration of PS-NPs (Fan et al., 2022). Exposure to PS-NPs disrupted glycolipid metabolism in mice, and reactive oxygen species (ROS) level was identified as a potential key signaling molecule (Fan et al., 2024).

Although the relationship between the gut microbiome and liver metabolites remained unclear, there was still a strong interaction between the gut microbiota and the liver (Shi et al., 2022). Changes in gut bacteria might lead to increased intestinal permeability and endotoxin translocation, thereby inducing transcriptional activation of pro-inflammatory genes and cytokines in the liver (Safari and Gérard, 2019). The alterations in the composition and function of the gut microbiota have been associated with findings suggestive of common metabolic disorders, such as obesity, non-alcoholic liver disease, type 2 diabetes, heart metabolic disease, and malnutrition (Agus et al., 2021; Fan and Pedersen, 2021). Meanwhile, the gut microbiota composition contributed to affect mucus barrier properties (Jakobsson et al., 2015). Altered or ecological dysregulation of the intestinal microbiota may also lead to intestinal barrier dysfunction, ultimately increasing susceptibility to intestinal inflammatory diseases (Loh and Blaut, 2012). Therefore, future studies should further reflect on the metabolic scenarios of microplastic exposure and determine the possible health outcomes of MPs through the gut–liver axis.

In summary, these findings contribute to a more realistic and comprehensive assessment of the health hazards associated with PET-MPs contamination. However, some limitations remain in our study, such as insufficient size selection of MPs and insufficient precision regarding the mechanisms of specific bacterial interactions. As noted, microplastics are a complex and dynamic mixture of polymers, additives, and contaminants in humans and the environment. Therefore, interdisciplinary interactions are essential for conducting systematic and comprehensive studies to evaluate their combined effects and potential mechanisms, and thereby assess the potential risks to human health.

5. Conclusions

Herein, the effects of 10 μ m PET-MPs on the blood biochemistry, inflammatory factors, histopathology, intestinal barrier function, and the gut microbiota were analyzed in mice. The study findings indicated that PET-MPs could induce inflammatory reactions, and histopathologic damage in the colon, impair intestinal barrier function, and cause gut microbiota dysbiosis. The results obtained in this study provide basic data on the potential effects of PET-MPs exposure to terrestrial mammals, but further research is still urgently needed to assess its potential impact on human health.

CRediT authorship contribution statement

Sun Xiangying: Writing – original draft, Methodology, Investigation, Formal analysis. **Zhuang Yin:** Visualization, Validation, Data curation. **Wang Yubang:** Project administration, Funding acquisition. **Zhang Zhenbo:** Methodology, Investigation. **AN Lihui:** Writing – review & editing, Supervision. **Xu Qiujin:** Supervision, Project

administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2025.118119.

Data availability

Data will be made available on request.

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